Introduction to LaTeX

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Abstract

This document demonstrates the basic structure of a LaTeX document. It introduces document classes, packages, title information, sections, paragraphs, and mathematical expressions.

1 Introduction

LaTeX is a document preparation system used for academic writing, technical reports, research papers, and mathematical documents.

It is especially useful when a document contains equations, tables, references, and structured sections.

2 Basic Document Structure

A LaTeX source file has two main parts: the preamble and the document body.

2.1 The Preamble

The preamble appears before the `\begin{document}` command. It contains the document class, packages, and title information.

2.2 The Document Body

The document body begins with `\begin{document}` and ends with `\end{document}`. It contains the actual content of the document.

3 A Mathematical Expression

LaTeX can display mathematical expressions clearly.

The Pythagorean theorem is:

$$a^2 + b^2 = c^2$$

4 Conclusion

This example demonstrates the basic structure of a LaTeX document. More advanced features such as formatting, lists, tables, figures, and colors are covered in the other practical programs.